

Multiple Choice (10 marks)

1. Magnetic flux has the unit of
 - (a) Volt
 - (b) Weber
 - (c) Watt
 - (d) Ohm
 - (e) Joule
2. Use the derivative property of Laplace transforms to solve the differential equation $y' - y = e^{-x}$ where $y(0) = 0$ is the initial value of y .
 3. $\sinh x$
 4. $e^x - e^{-x}$
 4. $e^x + x$
 5. $x^2 e^x$
 6. $\cosh x$

The resistance in the circuit of the moving coil of a dynamometer wattmeter should be

- (a) Low.
- (b) Very low.
- (c) High.
- (d) Almost zero.

Potentiometer method of DC voltage measurement is more accurate than direct measurement using a voltmeter because

- (a) It does not load the circuit at all.
- (b) It uses a potentiometer, which is more accurate than a voltmeter.
- (c) It uses a more precise measuring tool than a voltmeter.
- (d) It uses centre zero galvanometer instead of voltmeter.
- (e) It uses an ammeter instead of a voltmeter.

Express the Boyle temperature of a van der Waals gas in terms of the constants a , b , and R . The final expression for T_B must contain only a , b , and R .

- (a) $T_B = (a / 2 bR)$
- (b) $T_B = (R / ab)$

(c) $T_B = (2a / bR)$

(d) $T_B = (aR^2 / b)$

(e) $T_B = (bR / a)$

True / False (10 marks)

Answer True or False

6. True or False: The sequence $y[n]$ satisfying the given difference equation is $y[n] = 6 - (1/2)^n - (1/3)^n$.
7. True or False: The determinant of matrix A is 0, indicating that the matrix is singular and does not have an inverse.
8. True or False: The Laplace transform of $f(t) = \sin(kt)$ is $k / (s^2 - k^2)$, which is incorrect.
9. True or False: Evaluating the vector function at the point $(-1, 1, 2)$ yields $-3 - e^{-1}$.
9. True or False: The initial value of $f(t) = e^{-2t}u(t)$ at $t = 0^+$ is 0, and the slope is -2 .

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the factors influencing power delivery in a roller chain system, and calculate the power output for a 3/4 in. pitch chain on a 34-tooth sprocket rotating at 500 rpm.
12. Given a mixture of N and O gases with equal weights and specified absolute velocities, analyze the system to calculate the mass and molar average velocities. Discuss the implications of your findings.

End of Paper